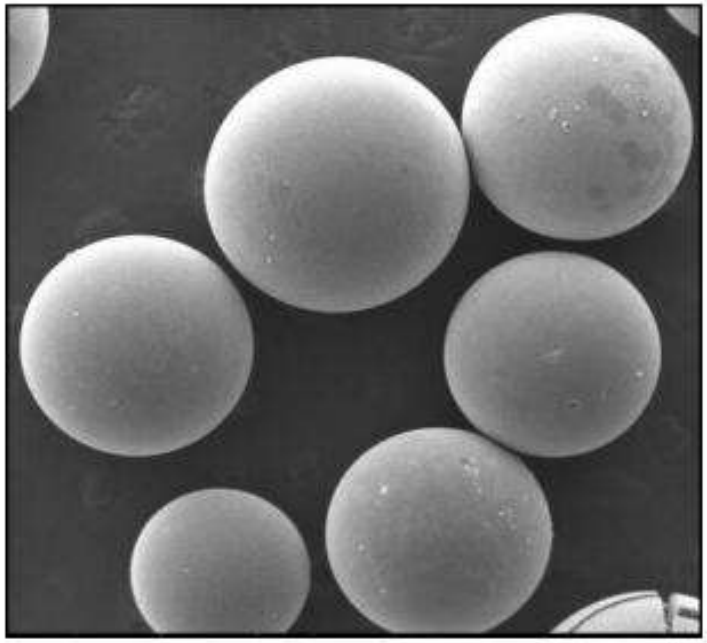
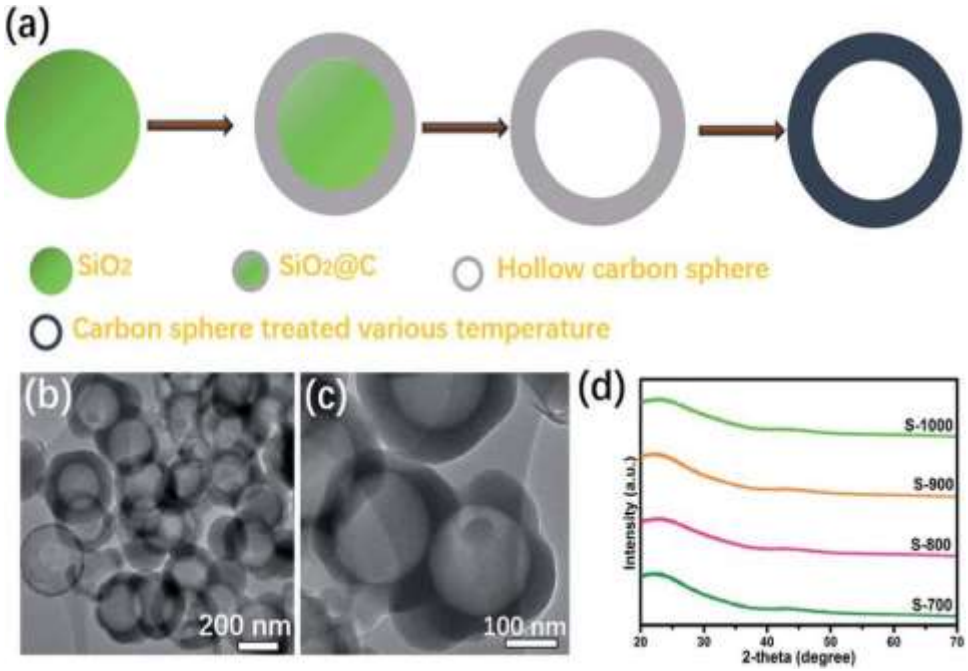
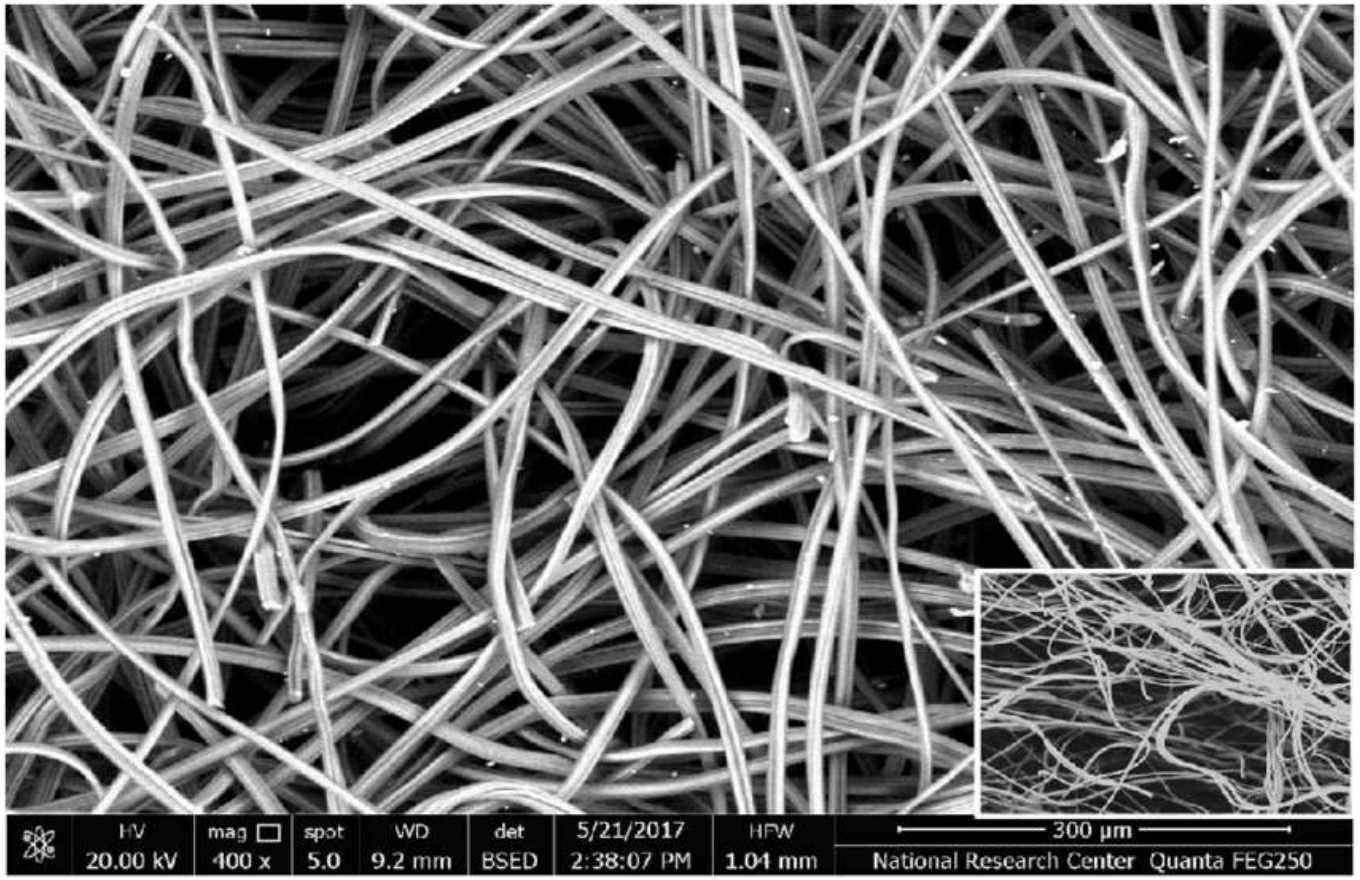
Types of carbon materials

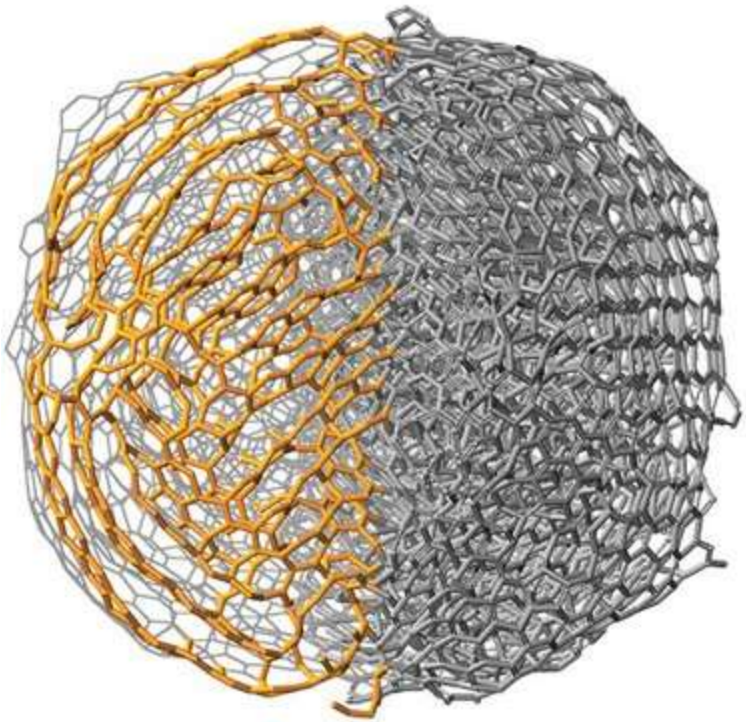
- Morphology
  - Carbon spheres
  - Carbon fibers
  - Carbonions
- Dimensional anisotropy
  - 0D - Fullerenes
  - 1D – Carbon nanotubes
  - 2D – Graphene
  - 2D - Graphdiyne
- Particle size
  - Carbon quantum dots





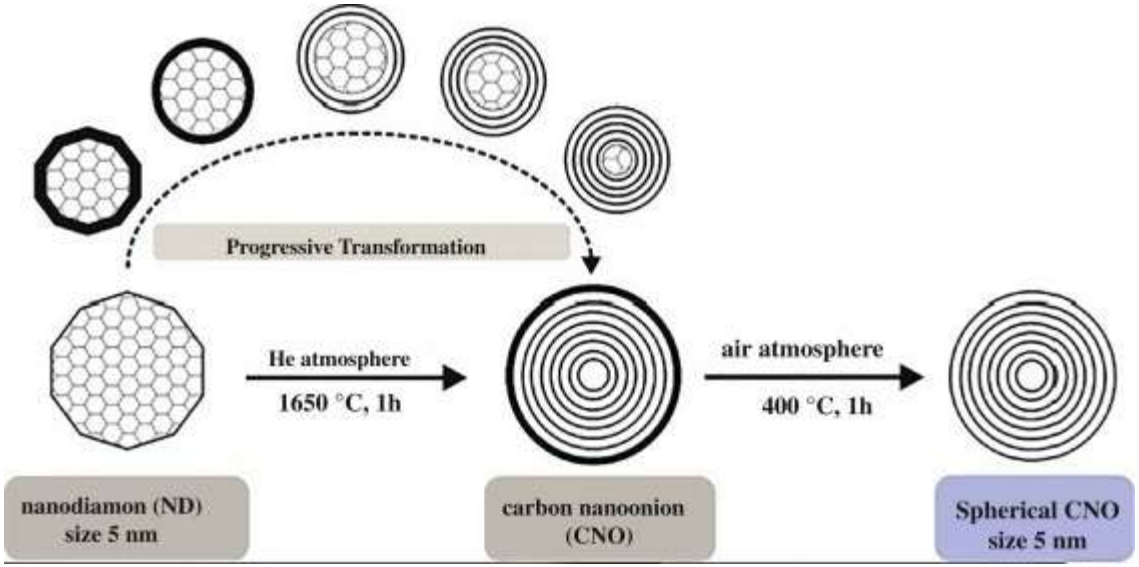
Prepared through template assisted method

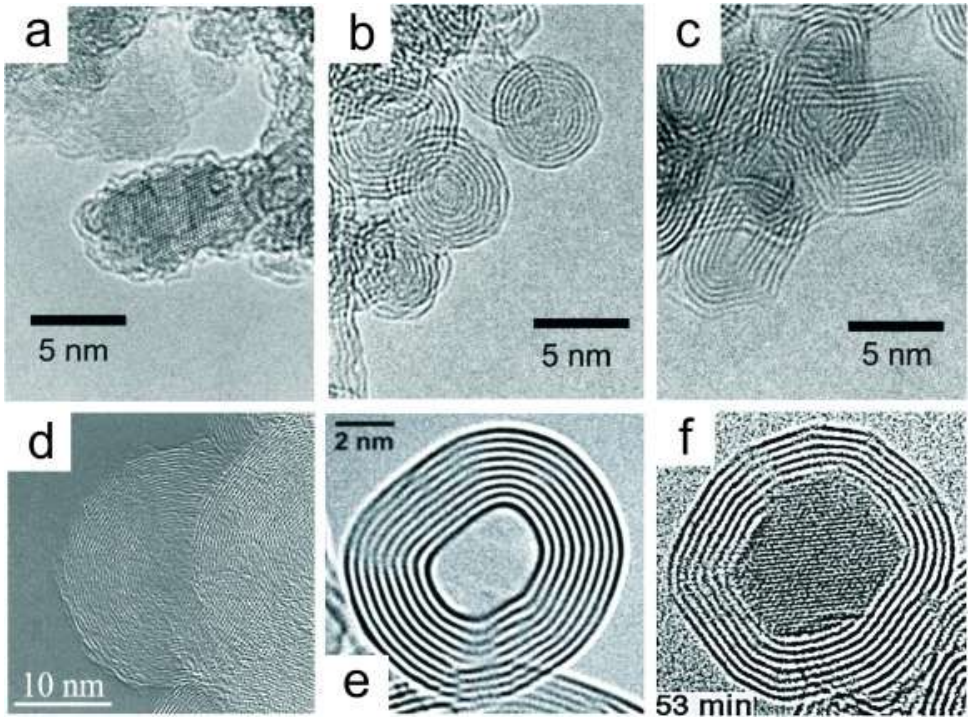

HV
20.00 kV
mag
400 x
spot
5.0
WD
9.2 mm
det
BSED
5/21/2017
2:38:07 PM
HFW
1.04 mm
300 µm
National Research Center Quanta FEG250

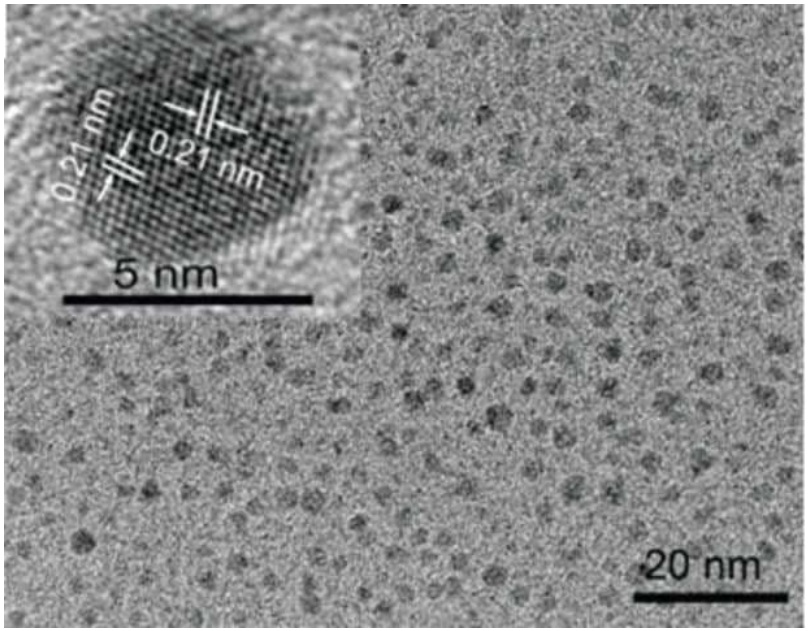


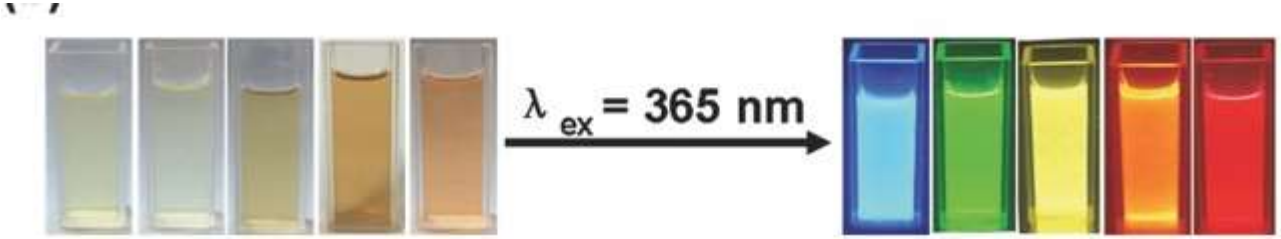
Carbon onion features:

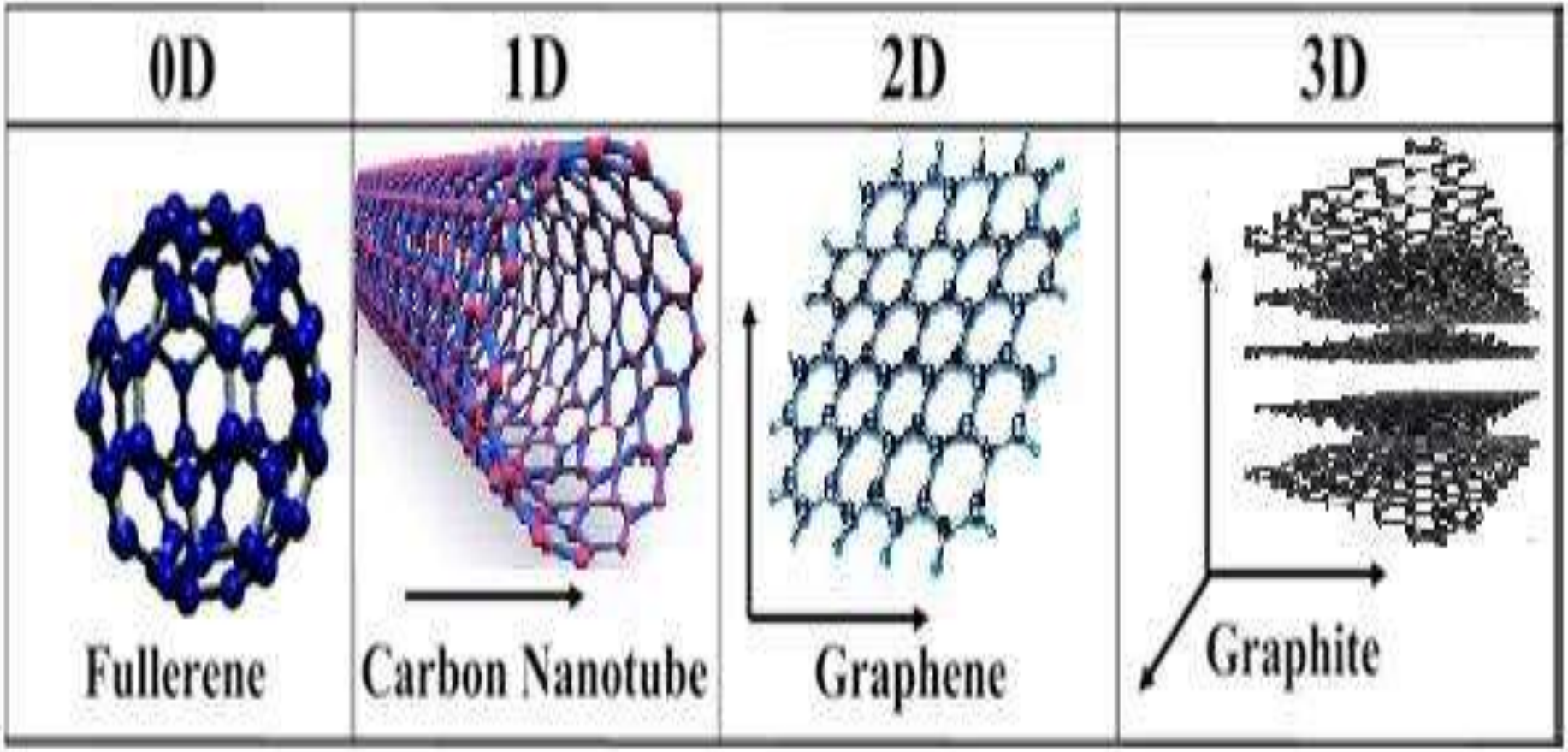
- Multiple shells of sp^2-hybridized carbon
- Varying degree of carbon ordering within the shell
- Shape varying from spherical to polyhedral
- Typical sizes varying between 5-10 nm (up to 100 nm possible)
- Low amount of heteroatoms
- Possible presence of non-sp^2-hybridized carbon
- Sometimes hollow core


Progressive Transformation
He atmosphere
1650 °C, 1h
air atmosphere
400 °C, 1h
nanodiamon (ND)
size 5 nm
carbon nanoonion
(CNO)
Spherical CNO
size 5 nm

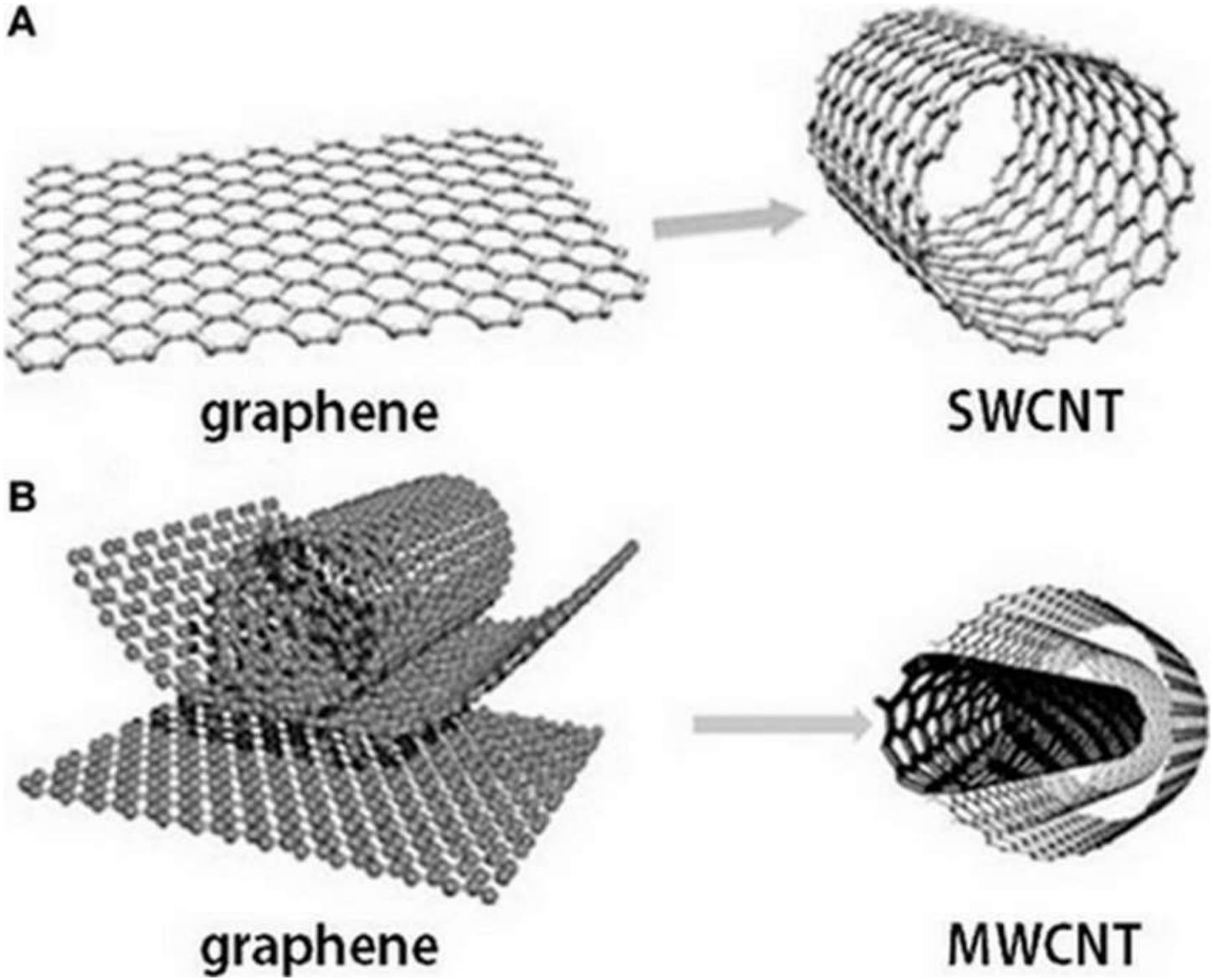

a
5 nm
b
5 nm
c
5 nm
d
10 nm
2 nm
e
f
53 min

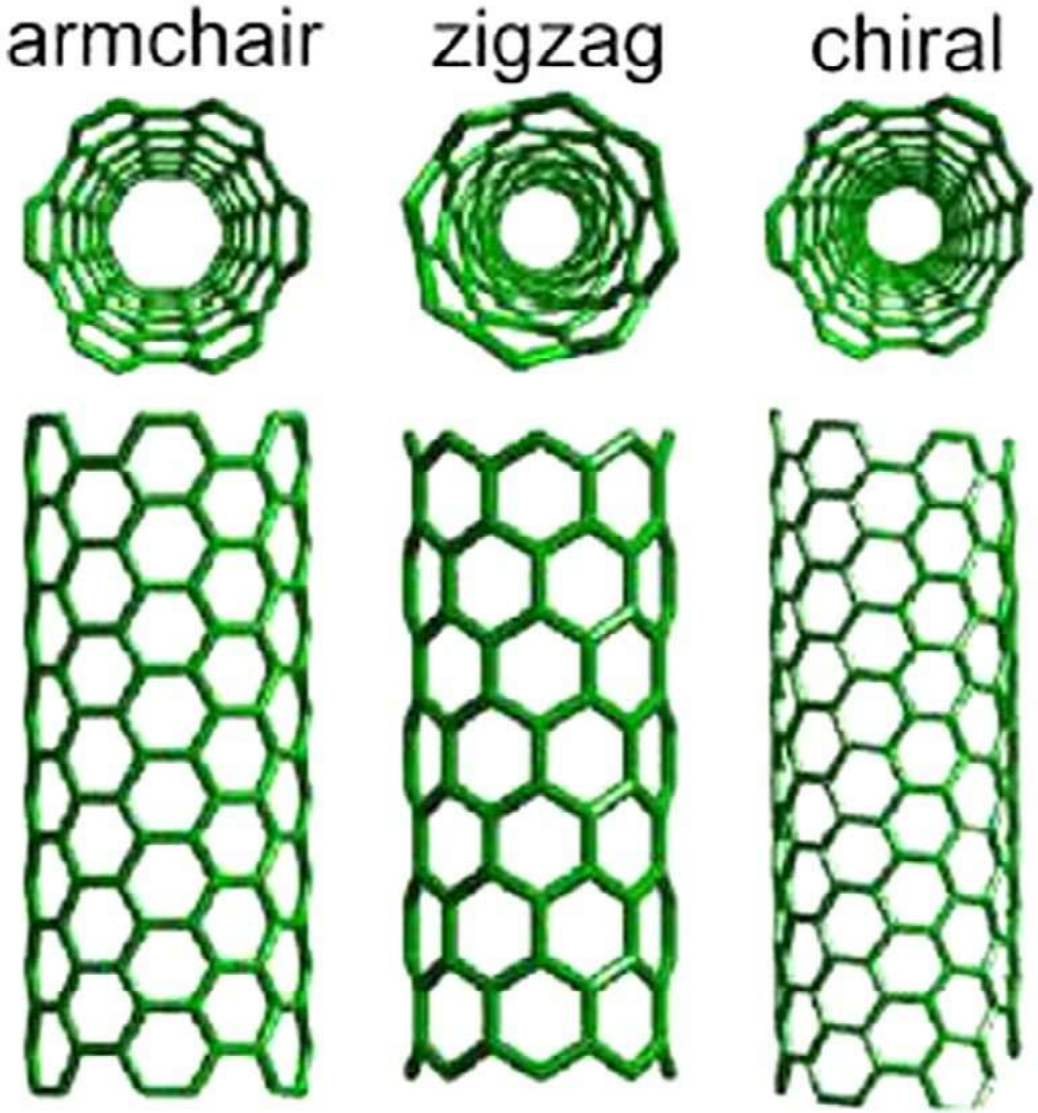

0.21 nm
0.21 nm
5 nm
20 nm


λ ex = 365 nm


0D
1D
2D
3D
Fullerene
Carbon Nanotube
Graphene
Graphite

Dimensional anisotropy



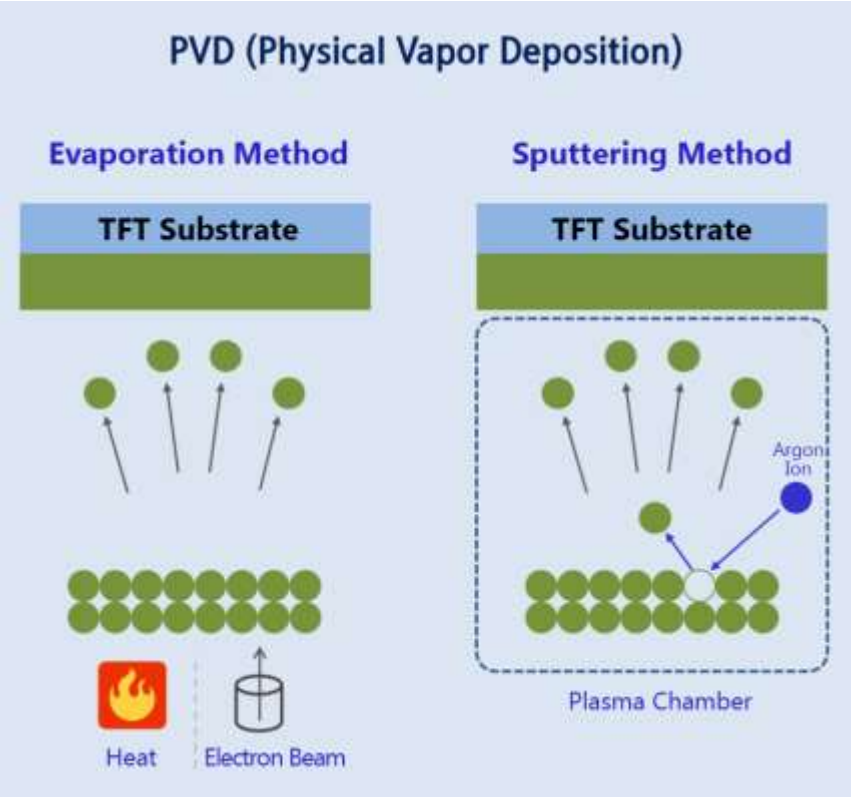

armchair
zigzag
chiral

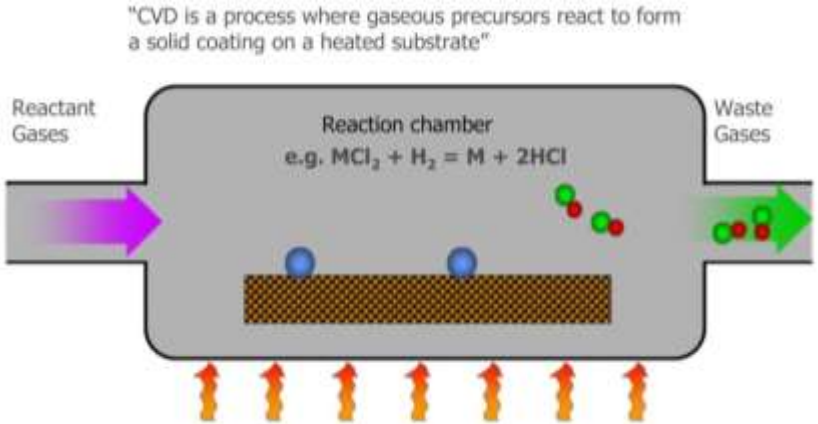
Vapor deposition methods

Physical vapor deposition

Chemical vapor deposition

Atmospheric pressure vapor CVD
Low pressure CVD
Plasma enhanced CVD
Ultra high vacuum CVD
Metal organic CVD

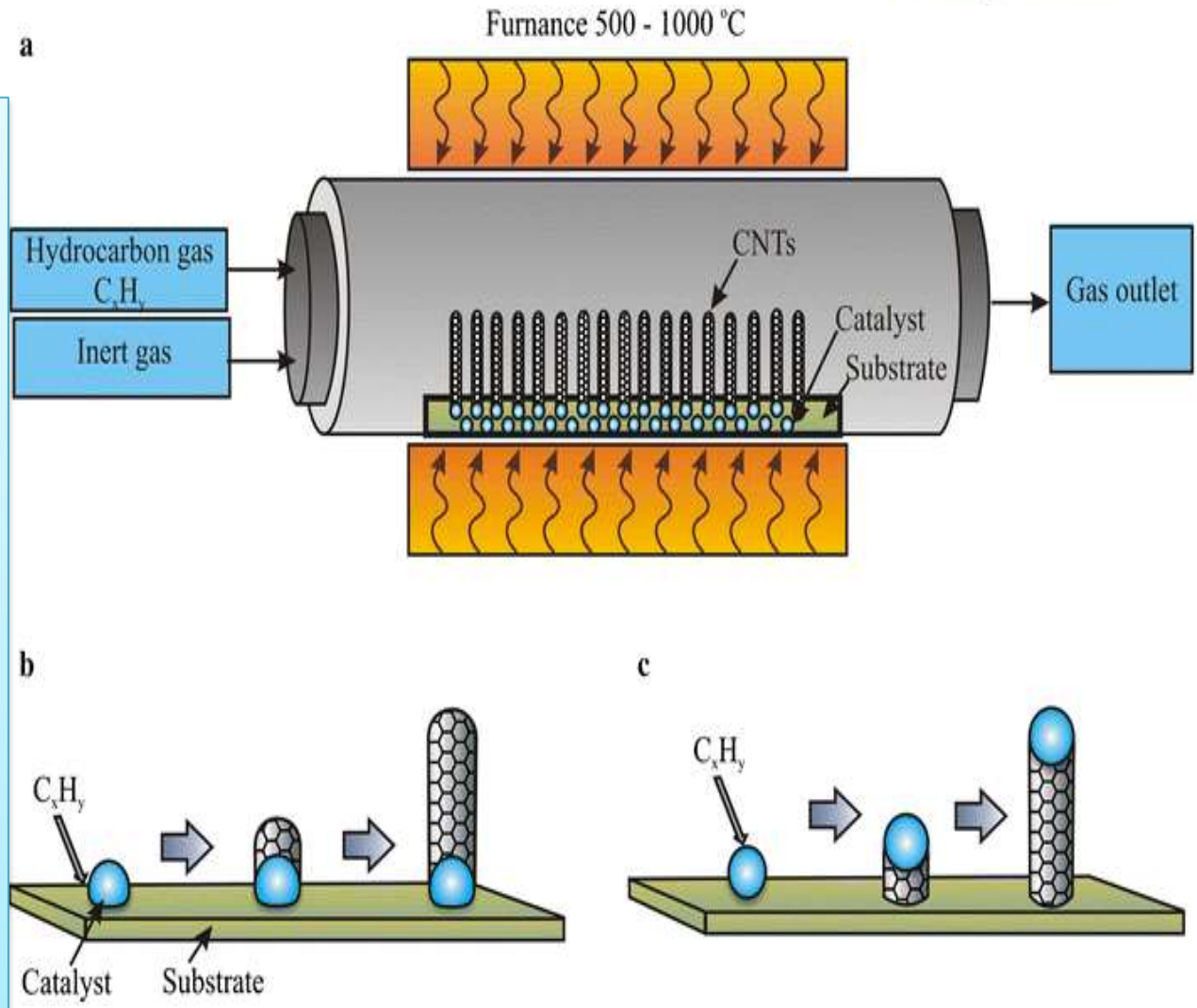




Synthesis of CNTs by modified CVD method:

Chemical Vapor Deposition: Requirements:

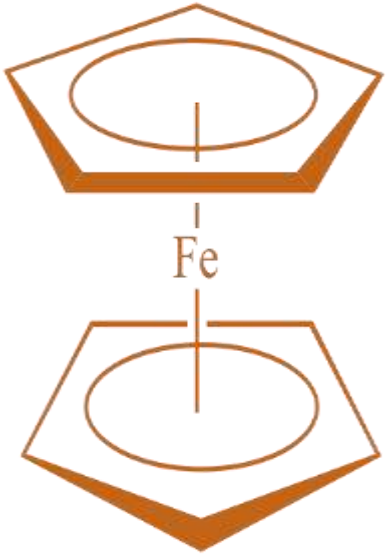
- High Temperature Tubular furnace (500 to 1500 ℃)
- Source of Carbon: Methane, ethylene, hydrocarbon gas, xylene, natural gas
- Substrate: carbon, quartz, silicon
- Inert gas: Argon, Hydrogen, Nitrogen
- Catalyst: Ferrocene, Nickellocene, Cobaltocene



Zaytseva, O., Neumann, G. Carbon nanomaterials: production, impact on plant lopment, agricultural and environmental applications. *Chem. Biol. Technol. Agric.* **3**, (2016). https://doi.org/10.1186/s40538-016-0070-8

Steps: Procedure involves as follows

- The cleaned quartz/carbon/silica substrate is placed in the middle of the tubular furnace.
- In order to maintain the inert atmosphere inside the furnace, initially the air/oxygen present inside the tubular furnace has to be removed by passing the argon or nitrogen gas for about 30 min.
- After maintain the inert gas atmosphere, the required temperature is set with slow heating rate.
- The hydrocarbon precursors in gaseous forms along with ferrocene and Benzene/toluene vapors are pumped into the reaction chamber.
- The furnace is heated up to 850–1000 °C and 550–700 °C for SWCNT and MWCNT production respectively.
- Initially, at high temperature, due the thermal decomposition of hydrocarbon, carbon atoms are formed are dissolved in the metal nanoparticle catalyst.
- Once the threshold concentration of carbon in the catalyst reached, a semi fullerene cap type of structure is formed due to precipitation of carbon atoms. This acts as a seed for further crystal growth, which further continues to grow in tubular form results in to CNT.
- After the formation of CNT on catalyst is completed as shown in the above figure, the supply of the reactant/catalyst varpors are stopped.
- Then the furnace temperature is reduced slowly to room temperature and supply of the inert gas also stopped.
- The CNT formed on catalyst is taken out along with the substrate, subjected to

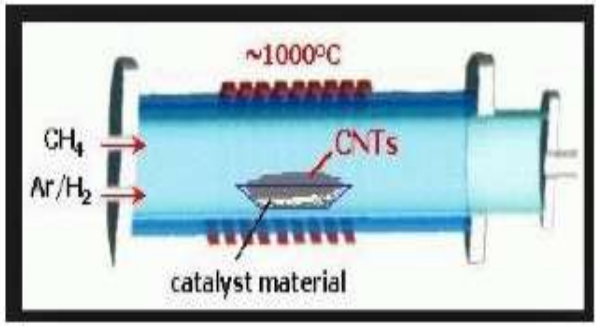

Fe

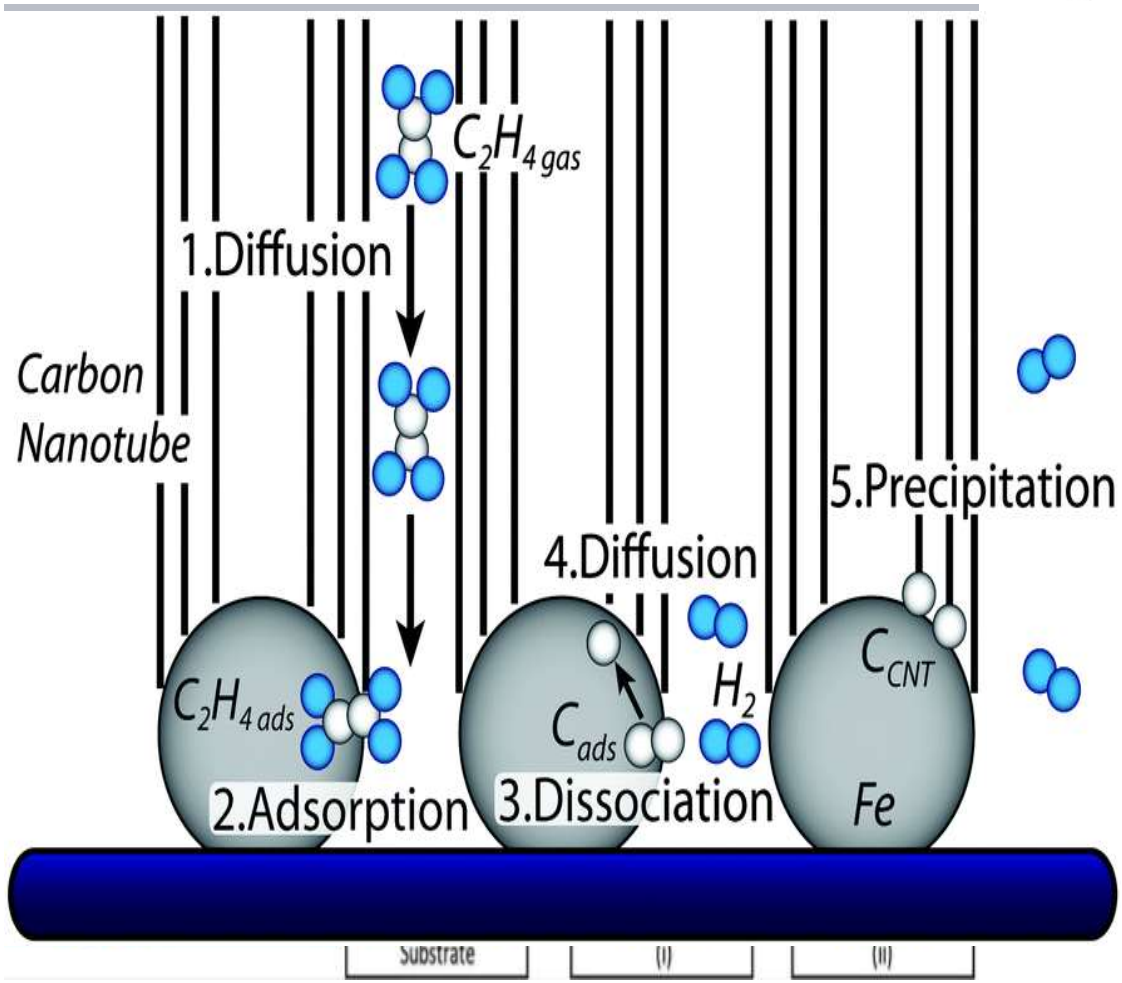
Multiwall Carbon
Nanotubes

1. Substrate is coated with metal seeds
2. Substrate is a glass or wafer surface and are exposed to various gaseous reactants/products
3. Gaseous precursors and carrier gases are transported into the deposition region
4. Adsorption of the reactant over the substrate surface
5. Surface reactions – Chemical decompositions and adsorptions
6. Desorption of byproducts away from the deposition region.
7. Interactions between the gaseous precursors and reactants with substrate determines the deposition rate

Metal seeds = Ni, Co, Fe

CNT growth mechanism in CVD





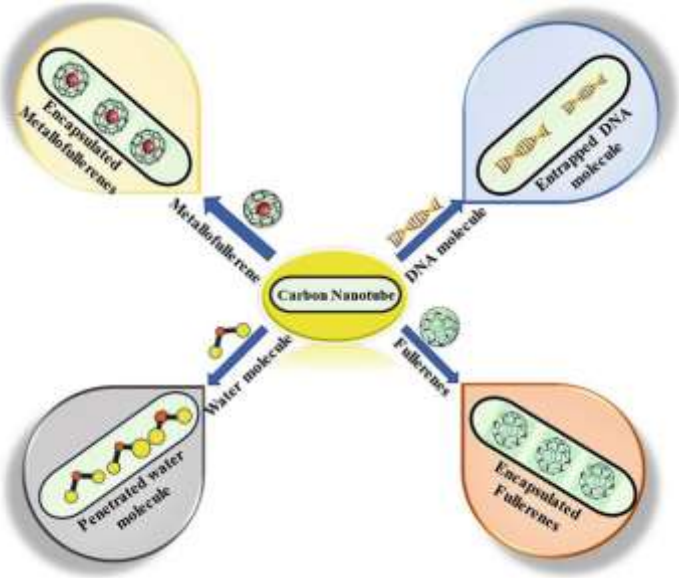
Functionalization of carbon nanotubes

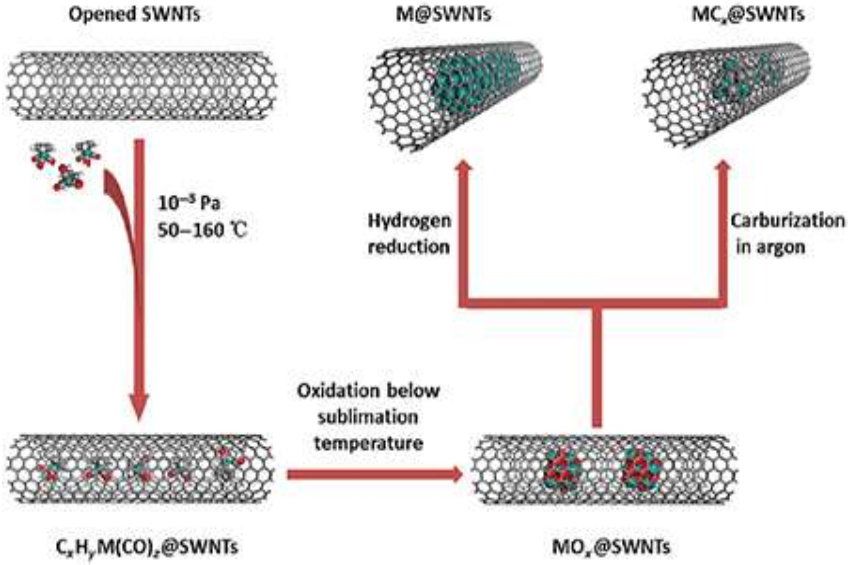
Hydrophobic and tend to agglomerate due to the Vander Waals forces – poor dispersion in liquid phase

Functionalization

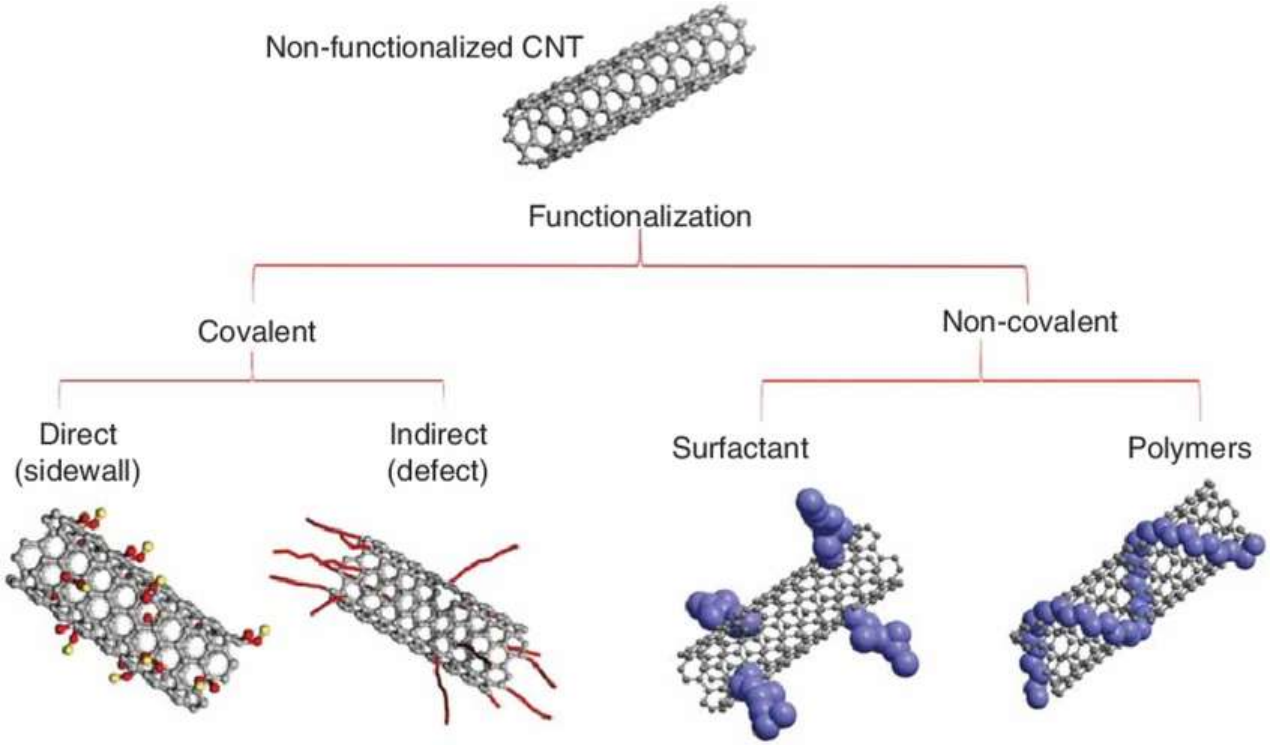
Hydrophilic and high degree of dispersion in various solvents

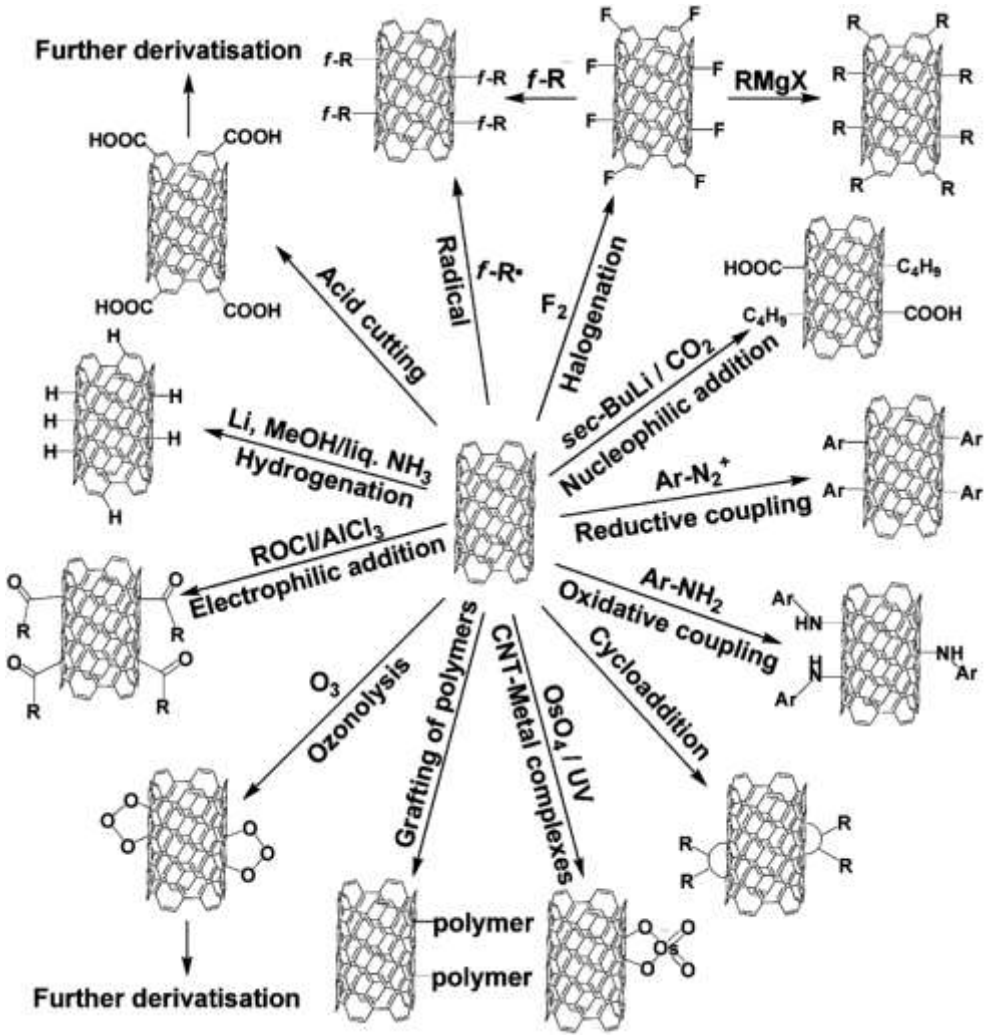
Endohedral Functionalization

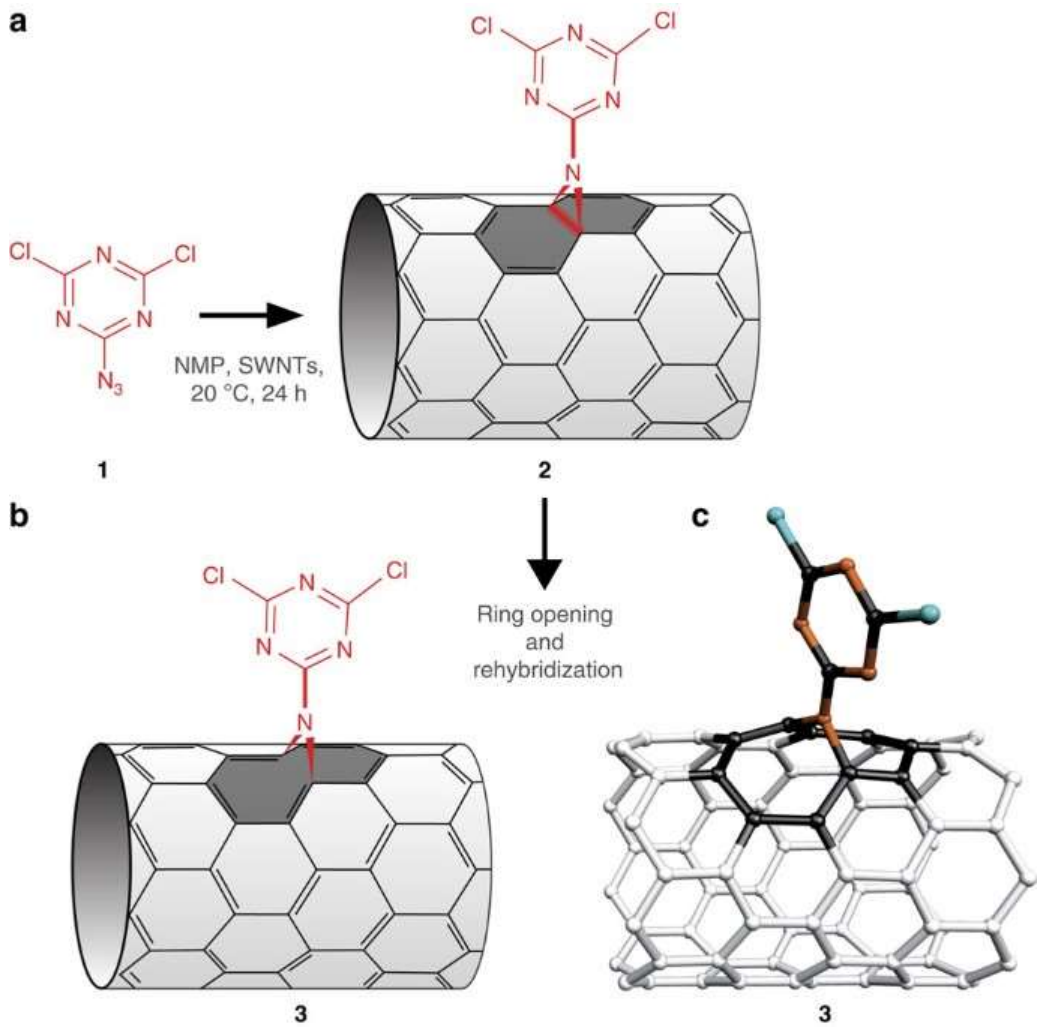


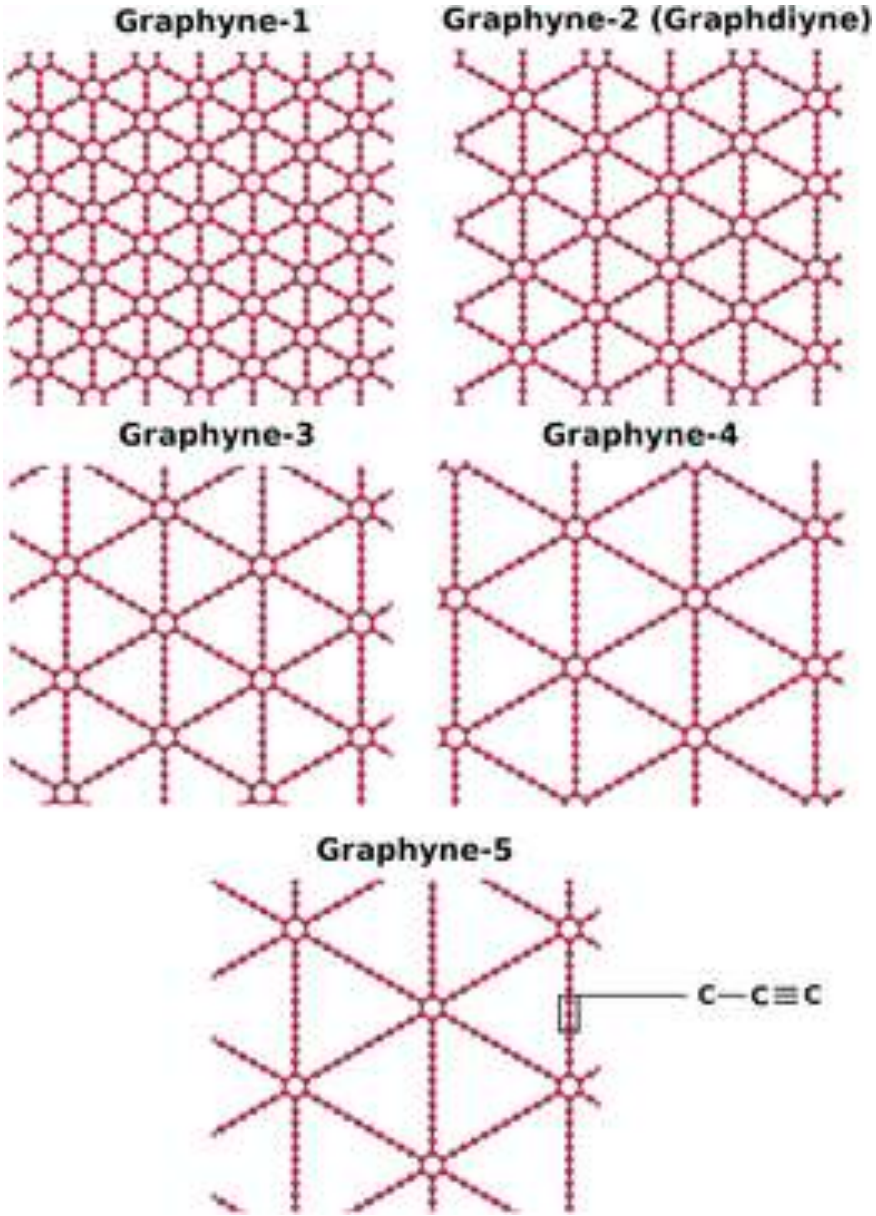


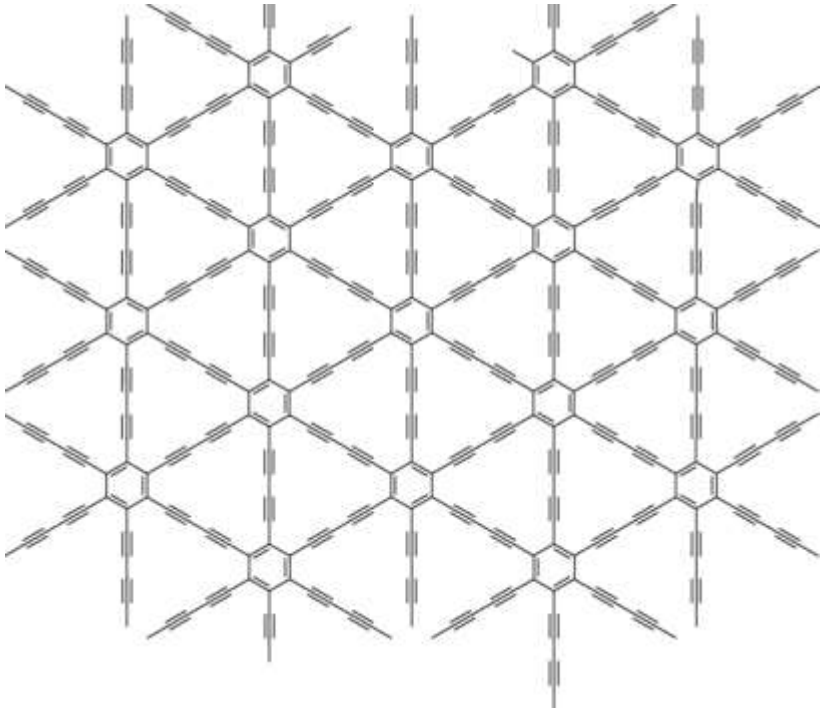
Exohedral Functionalization

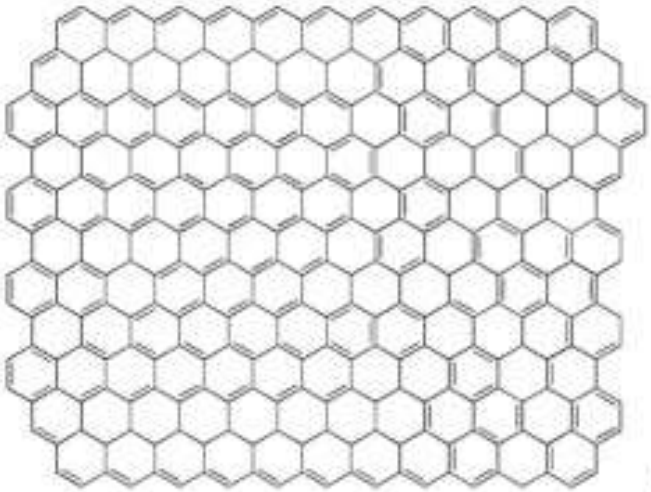



Further derivatisation
f-R
f-R
f-R
f-R
RMgX
HOOC
COOH
HOOC
COOH
Acid cutting
Radical
f-R•
F2
Halogenation
HOOC
C4H9
C4H9
COOH
sec-BuLi / CO2
Nucleophilic addition
Li, MeOH/liq. NH3
Hydrogenation
Ar-N2+
Reductive coupling
ROCl/AlCl3
Electrophilic addition
Ar-NH2
Oxidative coupling
Cycloaddition
O3
Ozonolysis
Grafting of polymers
CNT-Metal complexes
OsO4 / UV
polymer
polymer
Further derivatisation

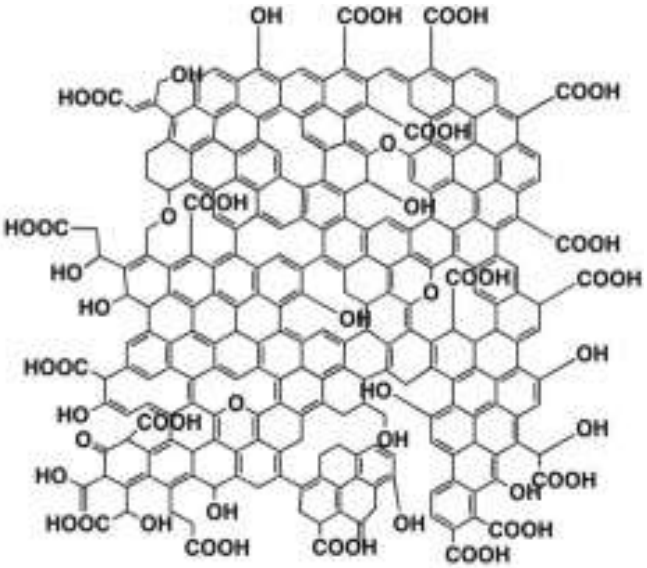

a
Cl
N
Cl
N
N
N
Cl
N
Cl
N
N
N
N3
NMP, SWNTs,
20 °C, 24 h
1
2
b
Cl
N
Cl
N
N
N
N
Ring opening
and
rehybridization
c
3
3


Graphyne-1
Graphyne-2 (Graphdiyne)
Graphyne-3
Graphyne-4
Graphyne-5
C—C≡C

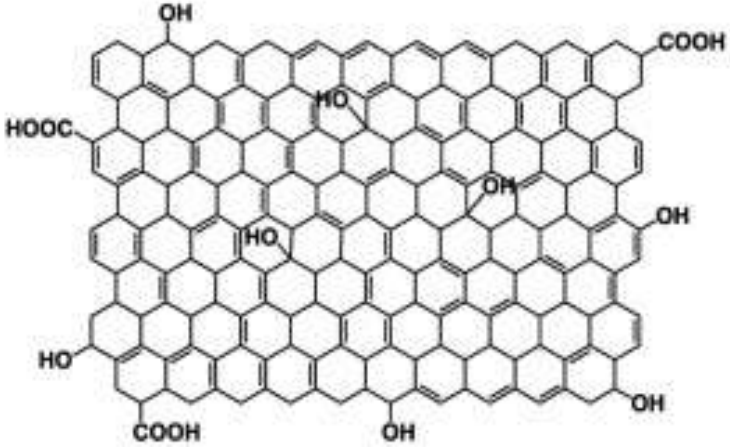




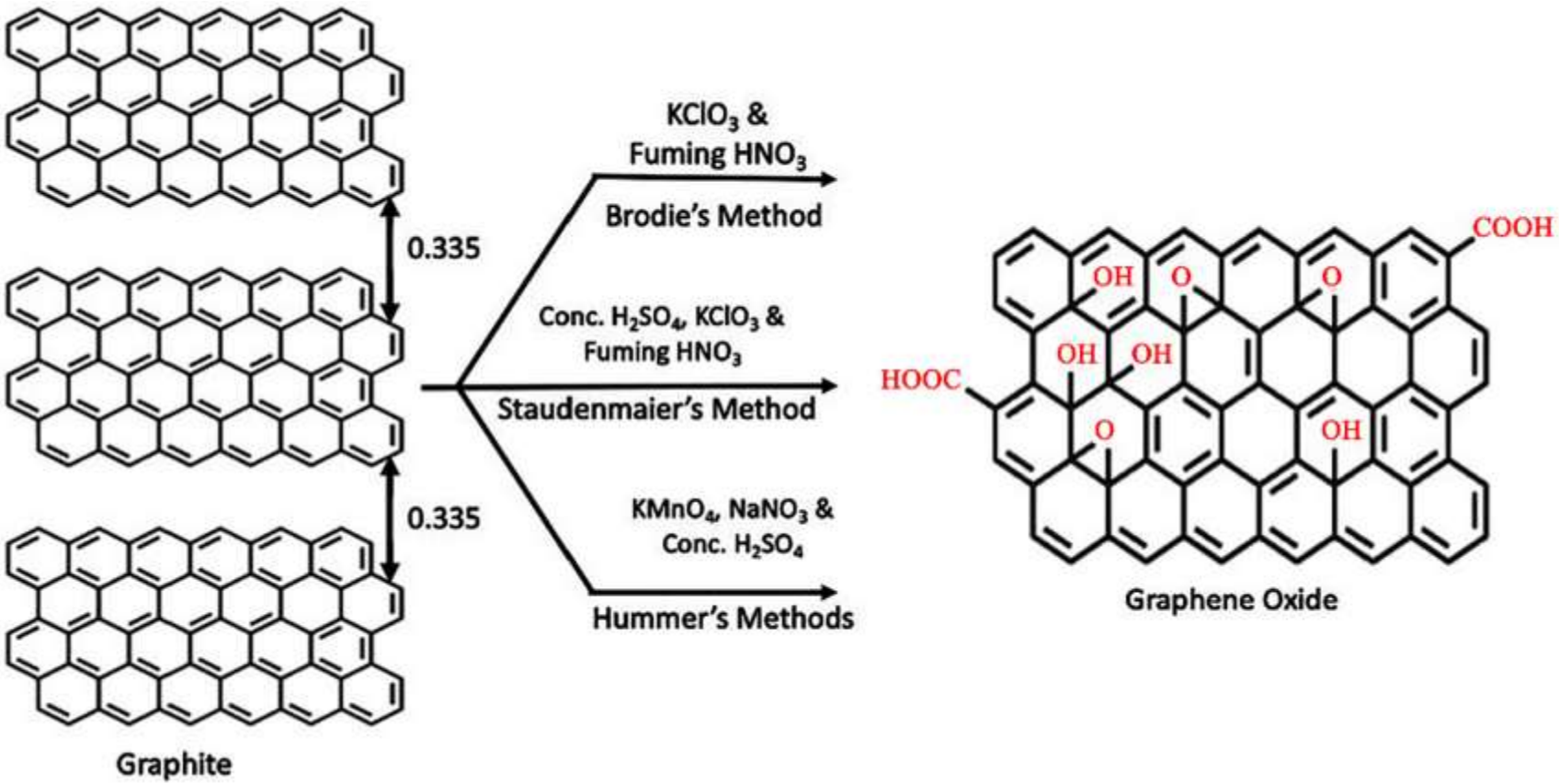
(A)


OH
COOH
COOH
HOOC
OH
COOH
COOH
O
COOH
OH
HOOC
O
COOH
HO
HO
O
COOH
COOH
OH
HOOC
OH
HO
HO
COOH
O
OH
OH
HO
OH
COOH
HOOC
OH
OH
COOH
OH
COOH
COOH
COOH

(B)


OH
COOH
HO
HOOC
OH
OH
HO
HO
OH
COOH
OH

(C)


0.335
0.335
Graphite
KClO3 &
Fuming HNO3
Brodie's Method
Conc. H2SO4, KClO3 &
Fuming HNO3
Staudenmaier's Method
KMnO4, NaNO3 &
Conc. H2SO4
Hummer's Methods
COOH
OH
O
O
OH
OH
HOOC
O
OH
Graphene Oxide

Stage-I: Synthesis of GO (graphene oxide) from graphite powder

Graphite powder (2 g) + 1 g sodium nitrate + 46 ml H_2SO_4 (dissolved with stirring in ice bath)

↓

Slow addition of $KMnO_4$ (6 g) during stirring of 30 minutes (temperature remains below 20°C)

↓

Stir for 30 minutes at 40°C and add 80 ml of water

↓

Stir for another 90 minutes at 90°C and add 200 ml water and 6 ml H_2O_2

↓

Filter and wash the solution with DI water and HCl

↓

Dissolve filtered paste in 100 ml water and ultrasonicate for 1 hour

↓

Centrifuge the solution and dry the obtained GO powder at room temperature

Stage-II: Synthesis of Graphene from GO

GO (100 mg) dissolved in to 100 ml water

↓

Add Hydrazine hydrate (10 µl, 0.2×10^{-3} mol)

↓

Reflux for one hour at 80°C

↓

Add sodium borohydride (1mg)

↓

Reflux for 36 hours at 100°C

↓

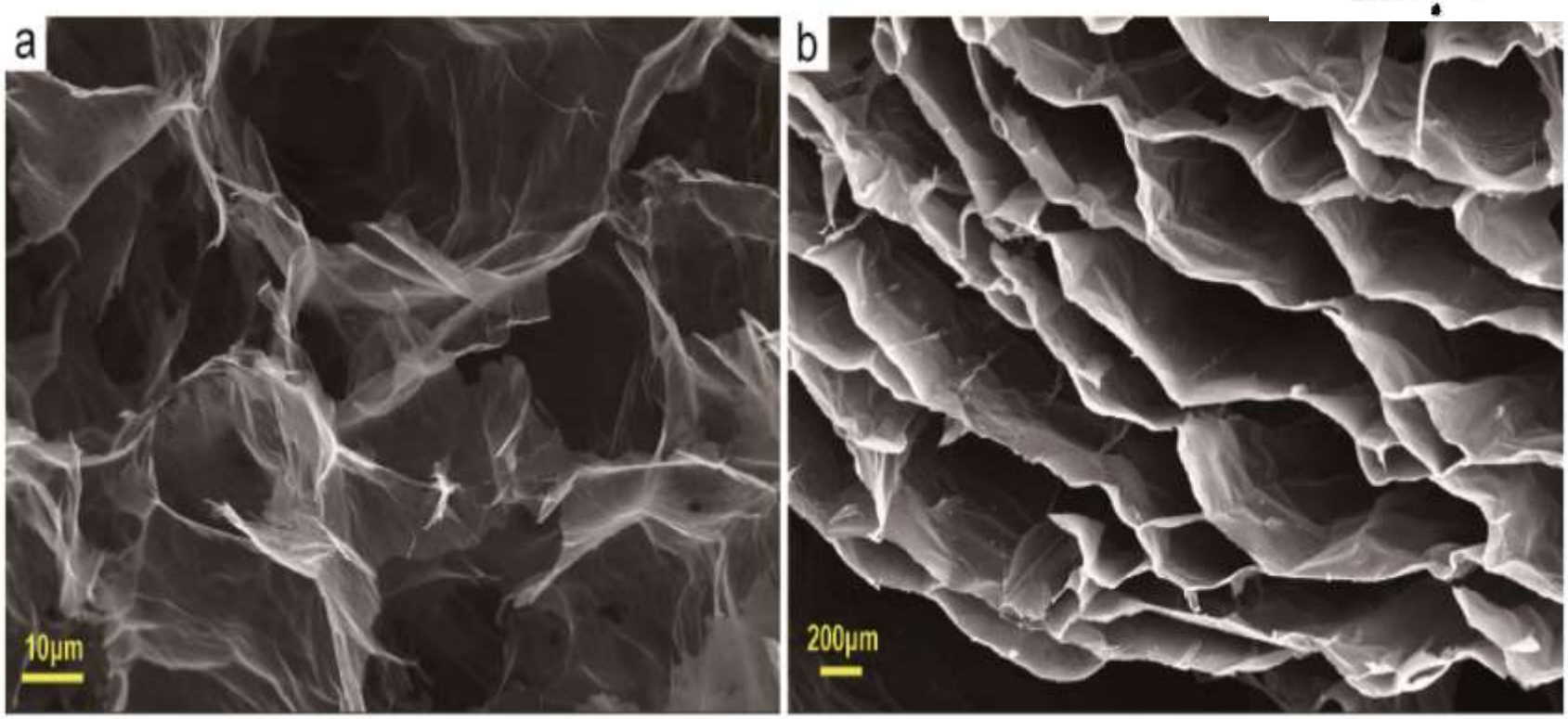
Filter and wash with DI water

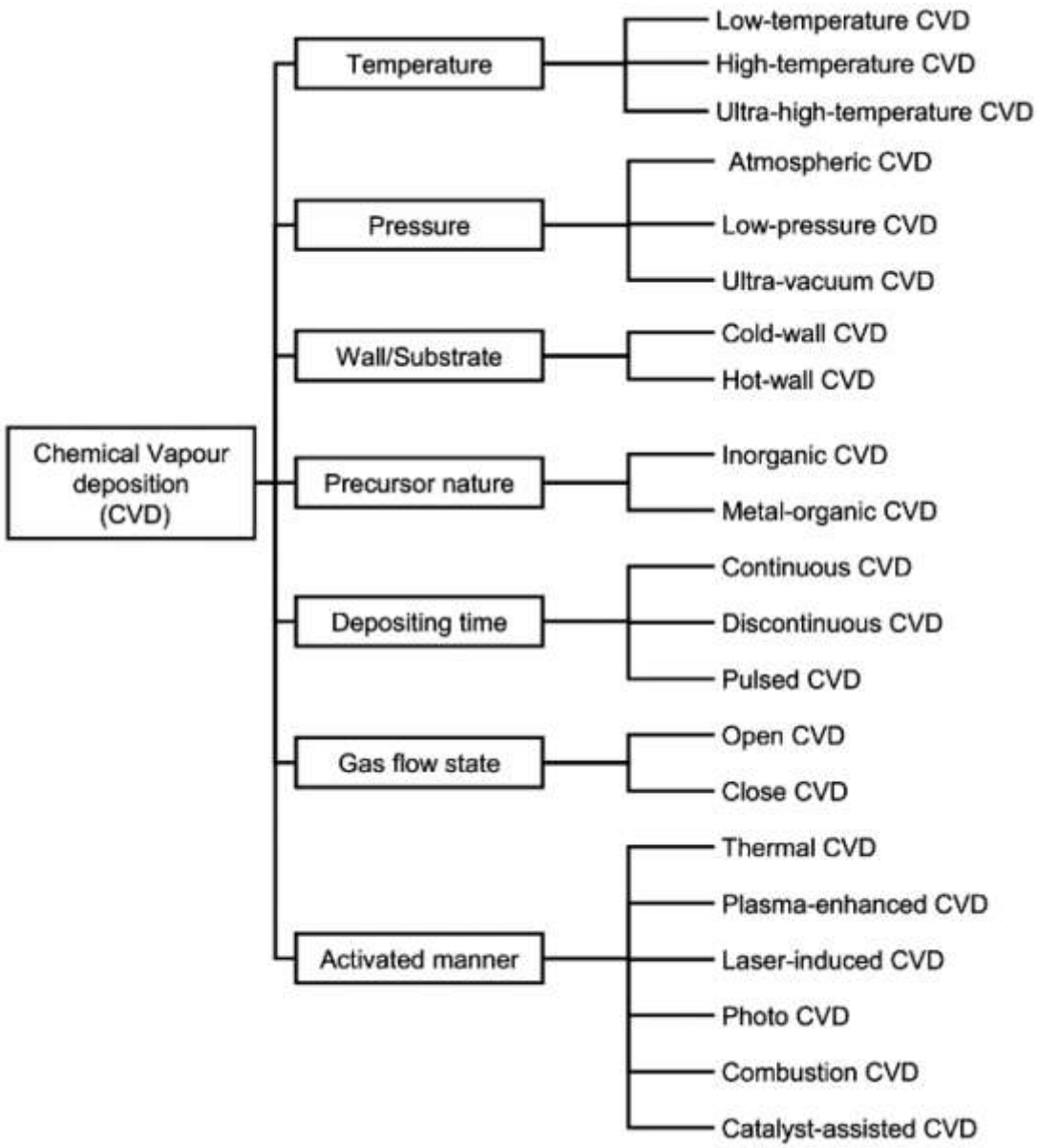
↓

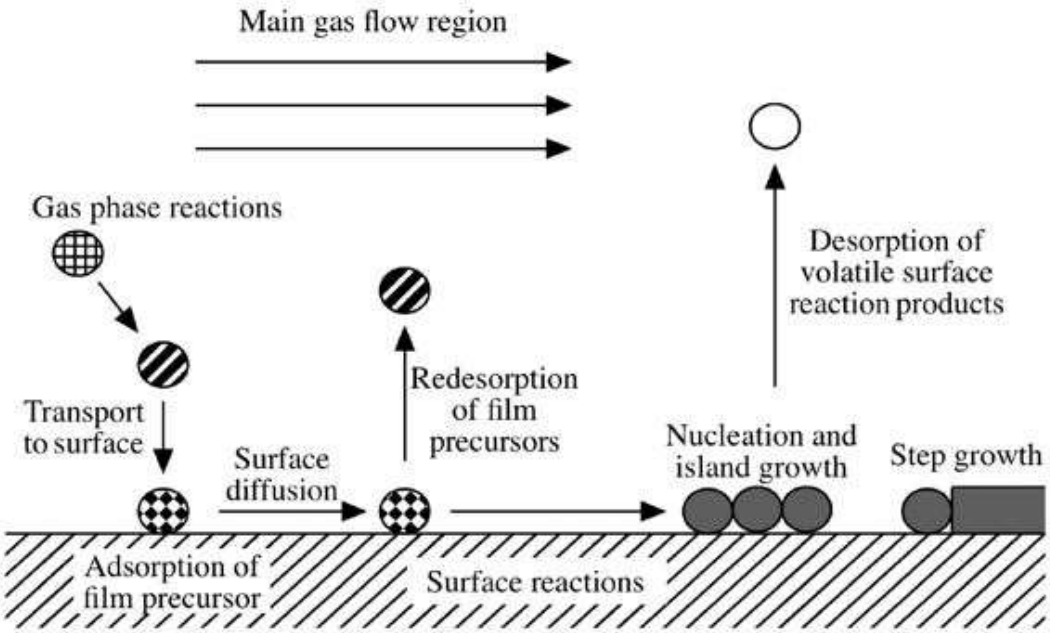
Dry at 60°C

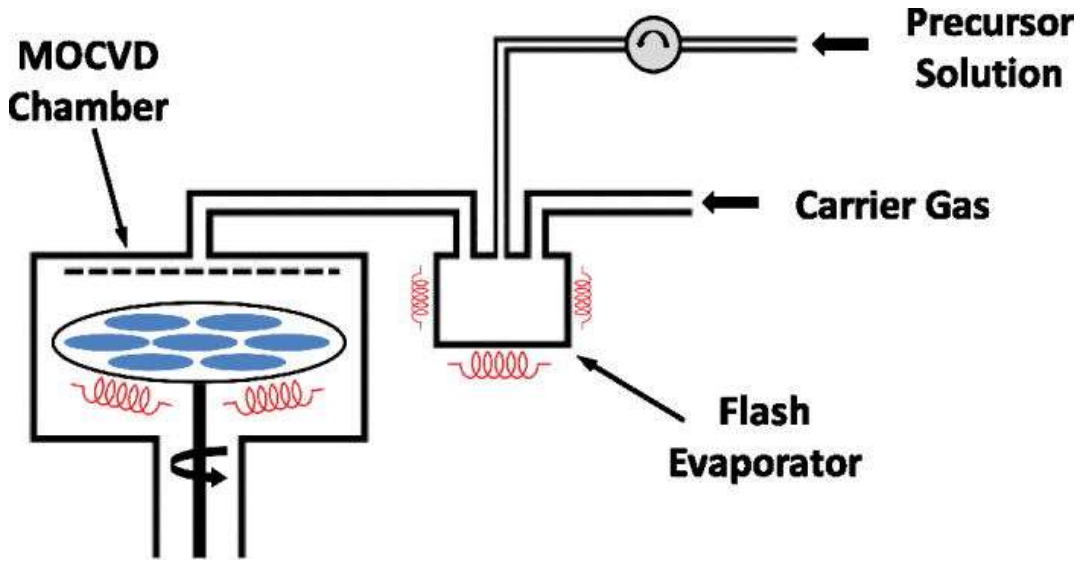
SEM images of Graphene

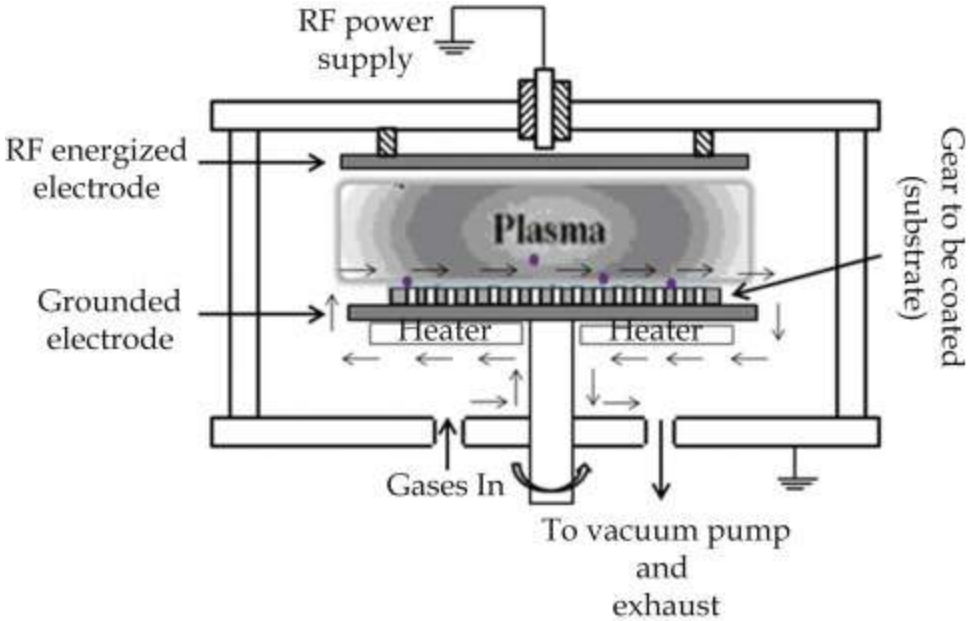
https://doi.org/10.3390/ma13153271

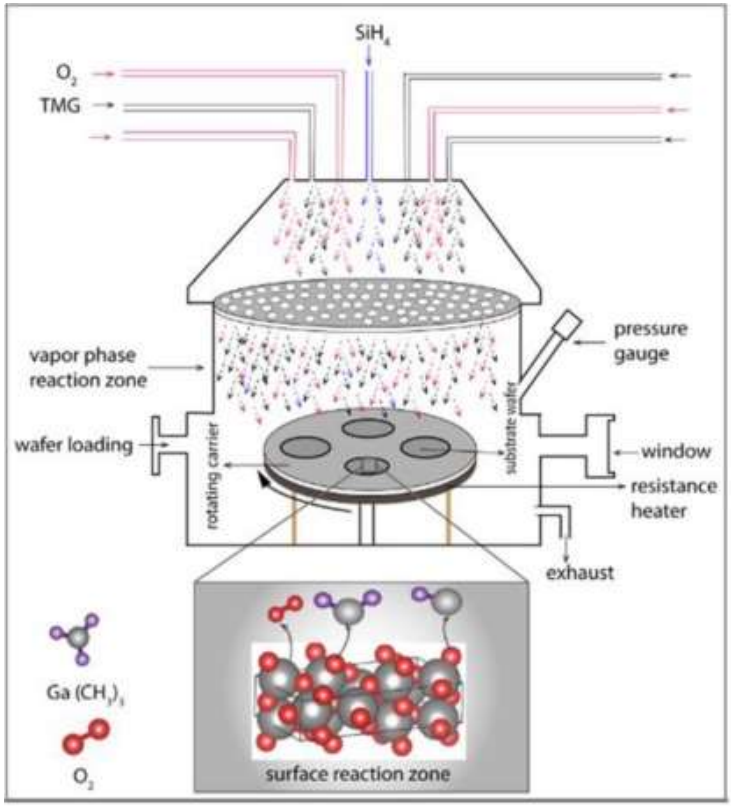



Chemical Vapour deposition (CVD)
Temperature
Low-temperature CVD
High-temperature CVD
Ultra-high-temperature CVD
Pressure
Atmospheric CVD
Low-pressure CVD
Ultra-vacuum CVD
Wall/Substrate
Cold-wall CVD
Hot-wall CVD
Precursor nature
Inorganic CVD
Metal-organic CVD
Depositing time
Continuous CVD
Discontinuous CVD
Pulsed CVD
Gas flow state
Open CVD
Close CVD
Activated manner
Thermal CVD
Plasma-enhanced CVD
Laser-induced CVD
Photo CVD
Combustion CVD
Catalyst-assisted CVD


Main gas flow region
Gas phase reactions
Transport to surface
Surface diffusion
Redesorption of film precursors
Desorption of volatile surface reaction products
Nucleation and island growth
Step growth
Adsorption of film precursor
Surface reactions


MOCVD Chamber
Precursor Solution
Carrier Gas
Flash Evaporator


RF power supply
RF energized electrode
Plasma
Grounded electrode
Heater
Heater
Gear to be coated (substrate)
Gases In
To vacuum pump and exhaust


SiH4
O2
TMG
vapor phase reaction zone
pressure gauge
wafer loading
rotating carrier
substrate wafer
window
resistance heater
exhaust
Ga (CH3)3
O2
surface reaction zone

MOCVD	PECVD
Operates at higher temperature	Operates at lower temperature
Thermal energy is required for activation process	Plasma energy triggers the reactions
Should be operated for those substrates that are thermally stable	Can be operated for any substrates (polymers)
Cleaning the chambers after the reaction is tedious	Flushing with nitrogen gas would be sufficient
Film growth is slow process	Film growth is rapid
Uniform thick film deposition is difficult to obtain, owing to high temperature	Very thick film can be attained at lower temperature
Difficult to obtain as high temperature leads to metal oxidation	Metals can be conveniently deposited